

 Marwadi University	Marwadi University Faculty of Technology Department of Information and Communication Technology	
Subject: Digital Signal and Image Processing (01CT1513)	Aim : Compute and analyze the correlation between three audio tracks	
OPE - 3	Date: 31/07/2026	Enrolment No:92400133050

Github Link :

Aim : Compute and analyze the correlation between three audio tracks

- **Cross-Correlation** Cross-correlation is used to compare two different signals. It helps determine whether the signals contain similar information and whether one signal is shifted with respect to the other. A higher peak in the cross-correlation graph indicates greater similarity between the two signals.

- **Auto-Correlation** Auto-correlation compares a signal with itself. It is mainly used to identify repeating patterns or periodicity within the signal. The highest value of auto-correlation always occurs at zero lag, because every signal is identical to itself. An audio signal is represented digitally as a sequence of amplitude samples. When two songs contain similar musical structures, their waveforms also become similar, resulting in a higher correlation. In this experiment: • **Original Song:** Contains vocals and instrumental music.

- **Karaoke Version:** Contains only instrumental music. • **Different Song:** Contains different melody, rhythm, tempo, and instruments. Because the original and karaoke versions share the same background music, they are expected to produce a higher correlation. On the other hand, the different song should produce a much lower correlation because its waveform pattern is unrelated.

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Technique: Normalized cross-correlation and autocorrelation were used to compare the original song, karaoke version, and a different song.

Results

- **Original vs Karaoke: Highest correlation indicating both tracks share nearly identical musical content.**

- **Original vs Different Song: Low correlation indicating unrelated audio.**

- **Karaoke vs Different Song: Low correlation for the same reason.**

- **Autocorrelation shows repeating patterns within each signal.**

Limitations

- **The accuracy of correlation analysis depends on the quality of the input audio signals.**
- **Background noise, recording distortions, and compression can affect the correlation values.**
- **Differences in signal length, sampling rate, or synchronization may reduce the measured similarity.**

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- **Correlation measures signal similarity only; it does not evaluate lyrics, vocals, or musical quality.**
- **Similar rhythms or melodies in different songs may produce moderate correlation values.**
- **The experiment is limited to three audio tracks, so the findings cannot be generalized to all music.**

Observations

- **After preprocessing, all audio signals were converted into a comparable digital format.**
- **The normalized cross-correlation between the original song and the karaoke version produced the highest peak among all comparisons.**
- **The correlation between the original song and the different song remained low throughout the analysis.**
- **The karaoke version and the different song also exhibited low correlation values.**
- **Autocorrelation of each signal showed repeating patterns corresponding to the periodic nature of audio signals.**
- **The correlation plots clearly distinguished similar audio tracks from unrelated ones.**

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Code :

```

import numpy as np

import librosa

import matplotlib.pyplot as plt

from scipy.signal import correlate

import os

original_file = r"/content/Starboy.mpeg"

karaoke_file = r"/content/Starboy - karaoke.mpeg"

different_file = r"/content/I wanna be Yours.mpeg"

for file in [original_file, karaoke_file, different_file]:

    if not os.path.exists(file):

        print("File not found:", file)

        exit()

sr = 22050

original, _ = librosa.load(original_file, sr=sr, mono=True)

karaoke, _ = librosa.load(karaoke_file, sr=sr, mono=True)

different, _ = librosa.load(different_file, sr=sr, mono=True)

length = min(len(original), len(karaoke), len(different))

original = original[:length]

karaoke = karaoke[:length]

different = different[:length]

```

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original = original / np.max(np.abs(original))
karaoke = karaoke / np.max(np.abs(karaoke))
different = different / np.max(np.abs(different))
def normalized_correlation(x, y):
    corr = correlate(x, y, mode="same")
    value = np.max(np.abs(corr)) / (np.linalg.norm(x) * np.linalg.norm(y))
    return corr, value
corr1, val1 = normalized_correlation(original, karaoke)
corr2, val2 = normalized_correlation(original, different)
corr3, val3 = normalized_correlation(karaoke, different)
auto_original = correlate(original, original, mode="same")
auto_karaoke = correlate(karaoke, karaoke, mode="same")
auto_different = correlate(different, different, mode="same")
plt.figure(figsize=(12,8))
plt.subplot(3,1,1)
plt.plot(original)
plt.title("Original Song - Starboy")
plt.xlabel("Samples")
plt.ylabel("Amplitude")
plt.subplot(3,1,2)
plt.plot(karaoke)

```

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plt.title("Karaoke Song - Starboy")
plt.xlabel("Samples")
plt.ylabel("Amplitude")
plt.subplot(3,1,3)
plt.plot(different)
plt.title("Different Song - i wanna be Yours")
plt.xlabel("Samples")
plt.ylabel("Amplitude")
plt.tight_layout()
plt.show()
plt.figure(figsize=(12,8))
plt.subplot(3,1,1)
plt.plot(auto_original)
plt.title("Autocorrelation - Original Song")
plt.xlabel("Lag")
plt.ylabel("Correlation")
plt.subplot(3,1,2)
plt.plot(auto_karaoke)
plt.title("Autocorrelation - Karaoke Song")
plt.xlabel("Lag")
plt.ylabel("Correlation")

```

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```

plt.subplot(3,1,3)
plt.plot(auto_different)
plt.title("Autocorrelation - Different Song")
plt.xlabel("Lag")
plt.ylabel("Correlation")
plt.tight_layout()
plt.show()
plt.figure(figsize=(12,8))
plt.subplot(3,1,1)
plt.plot(corr1)
plt.title("Cross Correlation - Original vs Karaoke")
plt.xlabel("Lag")
plt.ylabel("Correlation")
plt.subplot(3,1,2)
plt.plot(corr2)
plt.title("Cross Correlation - Original vs Different")
plt.xlabel("Lag")
plt.ylabel("Correlation")
plt.subplot(3,1,3)
plt.plot(corr3)
plt.title("Cross Correlation - Karaoke vs Different")

```

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```

plt.xlabel("Lag")
plt.ylabel("Correlation")
plt.tight_layout()
plt.show()

pairs = [
    "Original\nvs\nKaraoke",
    "Original\nvs\nDifferent",
    "Karaoke\nvs\nDifferent"
]

values = [val1, val2, val3]

plt.figure(figsize=(7,5))
plt.bar(pairs, values)
plt.title("Normalized Correlation Comparison")
plt.ylabel("Correlation Value")
plt.ylim(0,1)
plt.grid(axis="y")
plt.show()

print("\n=====")
print("Normalized Correlation Values")
print("=====")
print(f'Original vs Karaoke   : {val1:.4f}')

```


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```

print(f"Original vs Different : {val2:.4f}")
print(f"Karaoke vs Different : {val3:.4f}")
print("\nInterpretation")
if val1 > val2 and val1 > val3:
    print("- Original and Karaoke have the highest similarity.")
    print("- This is expected because both are from the same song.")
else:
    print("- Please verify the selected audio files.")
print("\nCorrelation Scale")
print("0.90 - 1.00 : Very High Similarity")
print("0.70 - 0.89 : High Similarity")
print("0.40 - 0.69 : Moderate Similarity")
print("0.00 - 0.39 : Low Similarity")

```

Output :



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Department of Information and Communication Technology

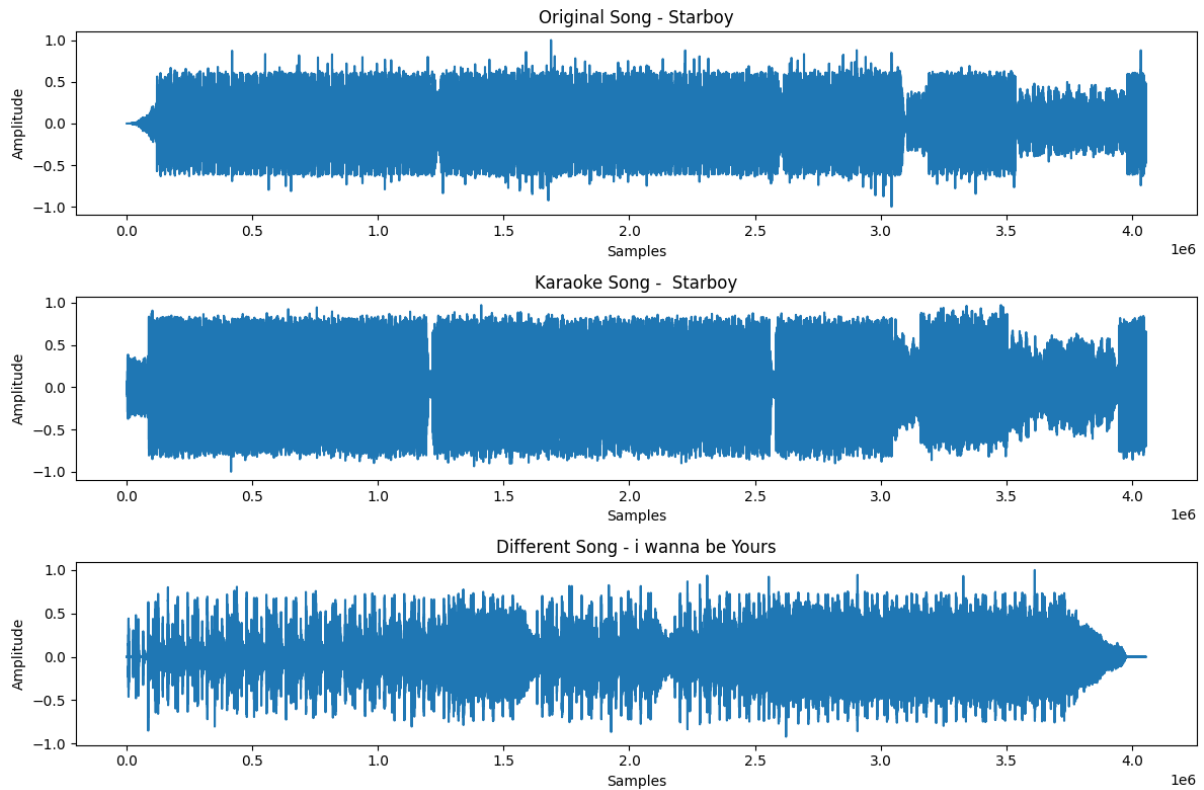
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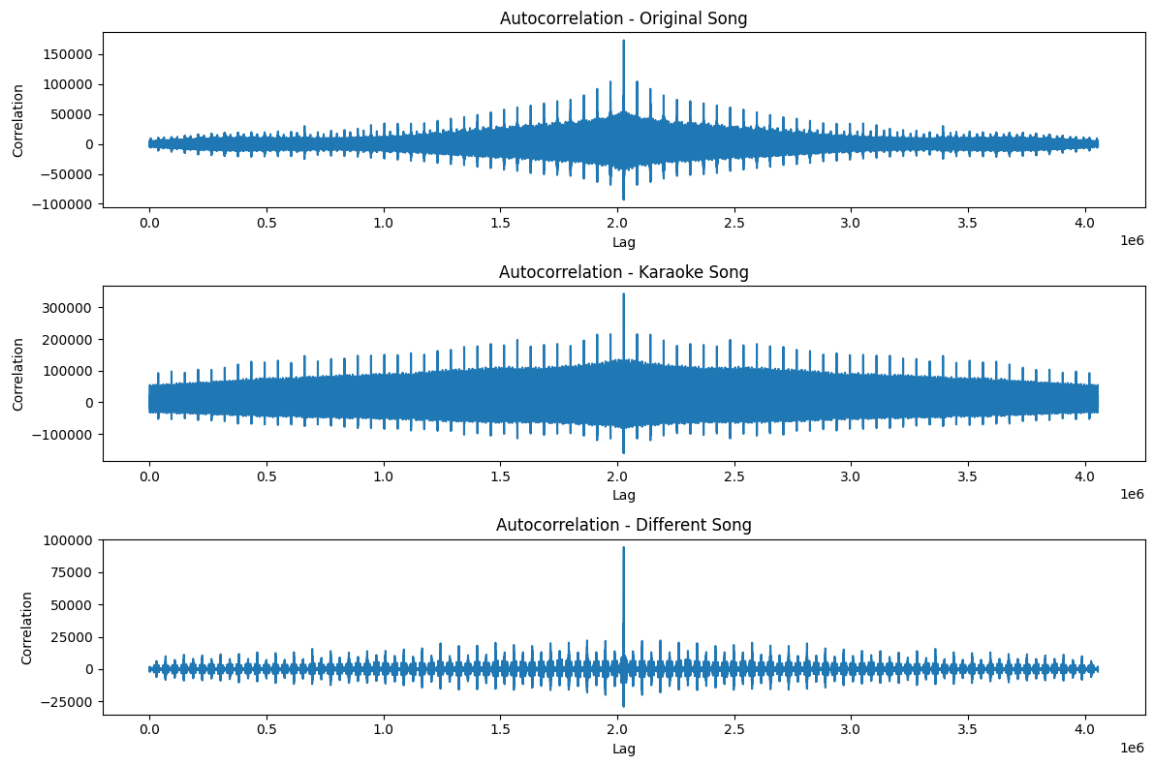
**Subject: Digital Signal and
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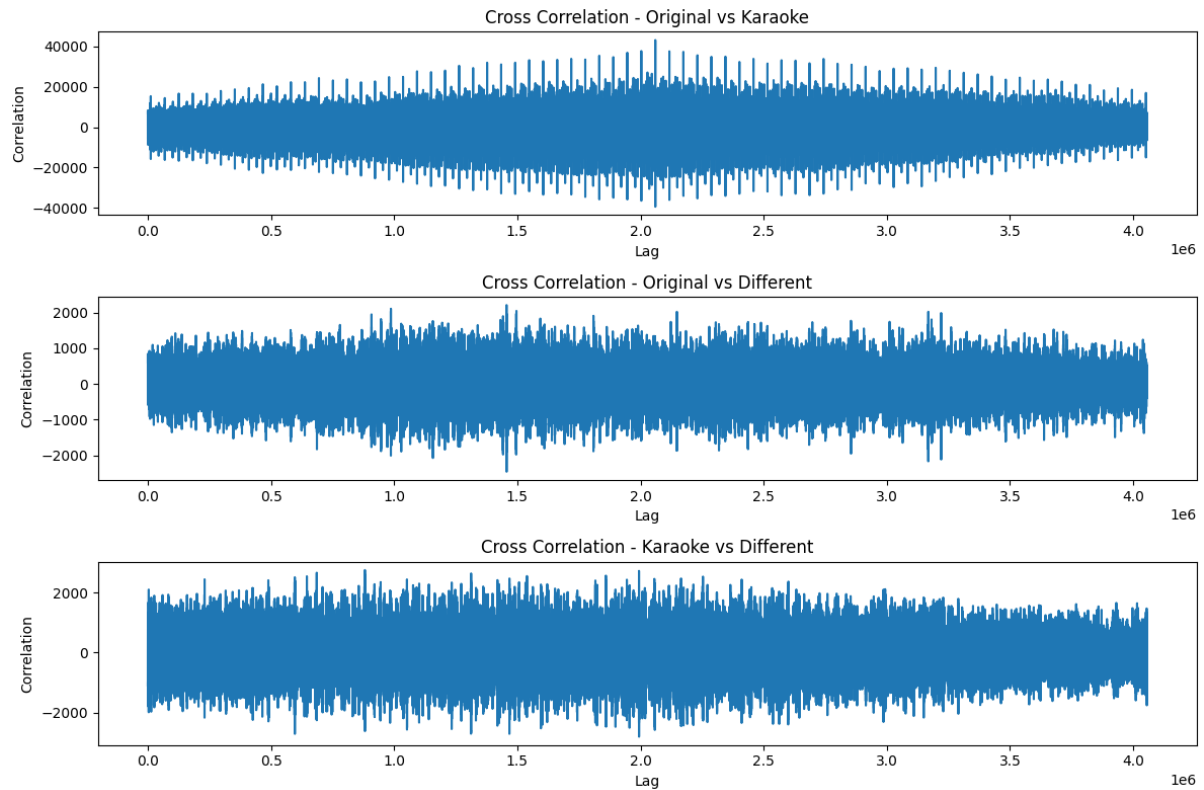
**Subject: Digital Signal and
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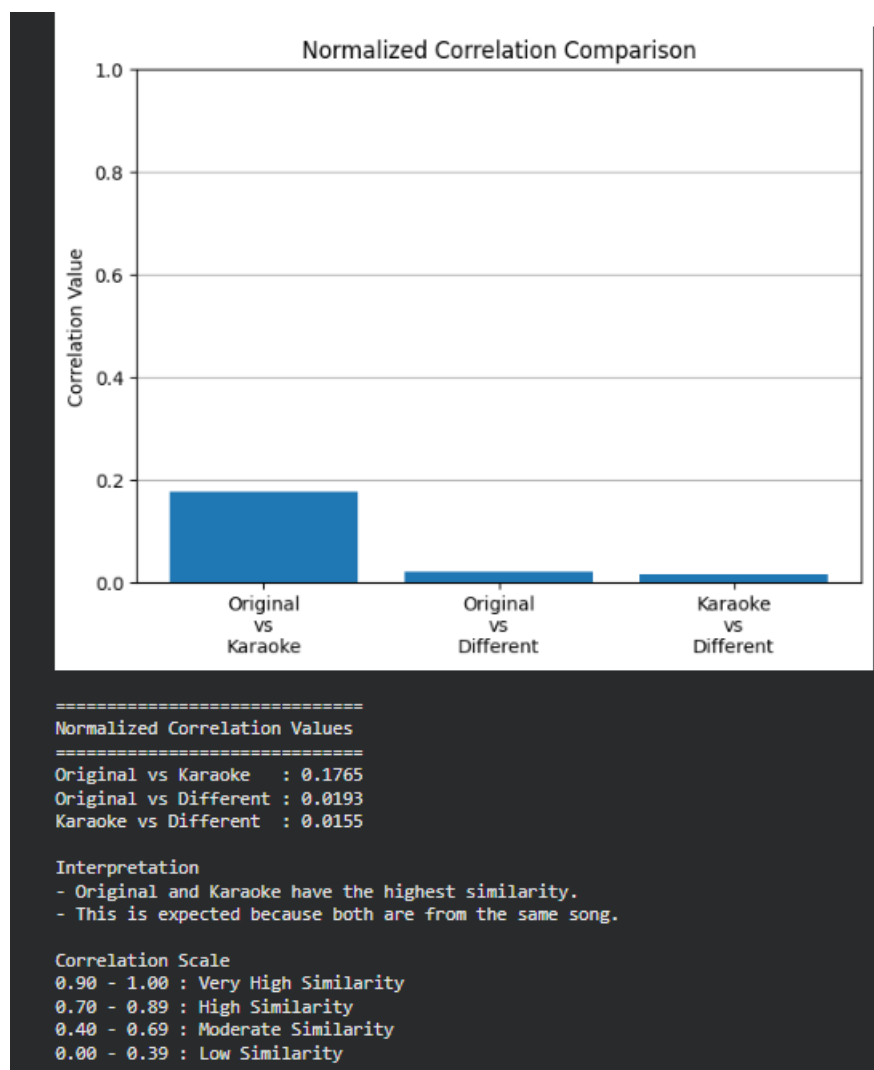
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Conclusion : The experiment successfully demonstrated the use of correlation techniques to compare the similarity between different audio signals. The normalized cross-correlation analysis showed that the original song and its karaoke version have the highest similarity because they share the same musical structure and rhythm, despite the absence of vocals in the karaoke track. In contrast, the original song and the completely different song, as well as the karaoke version and the different song, produced significantly lower correlation values, indicating minimal similarity between these signals.

The autocorrelation analysis further highlighted the periodic and repeating patterns present within each individual audio signal.